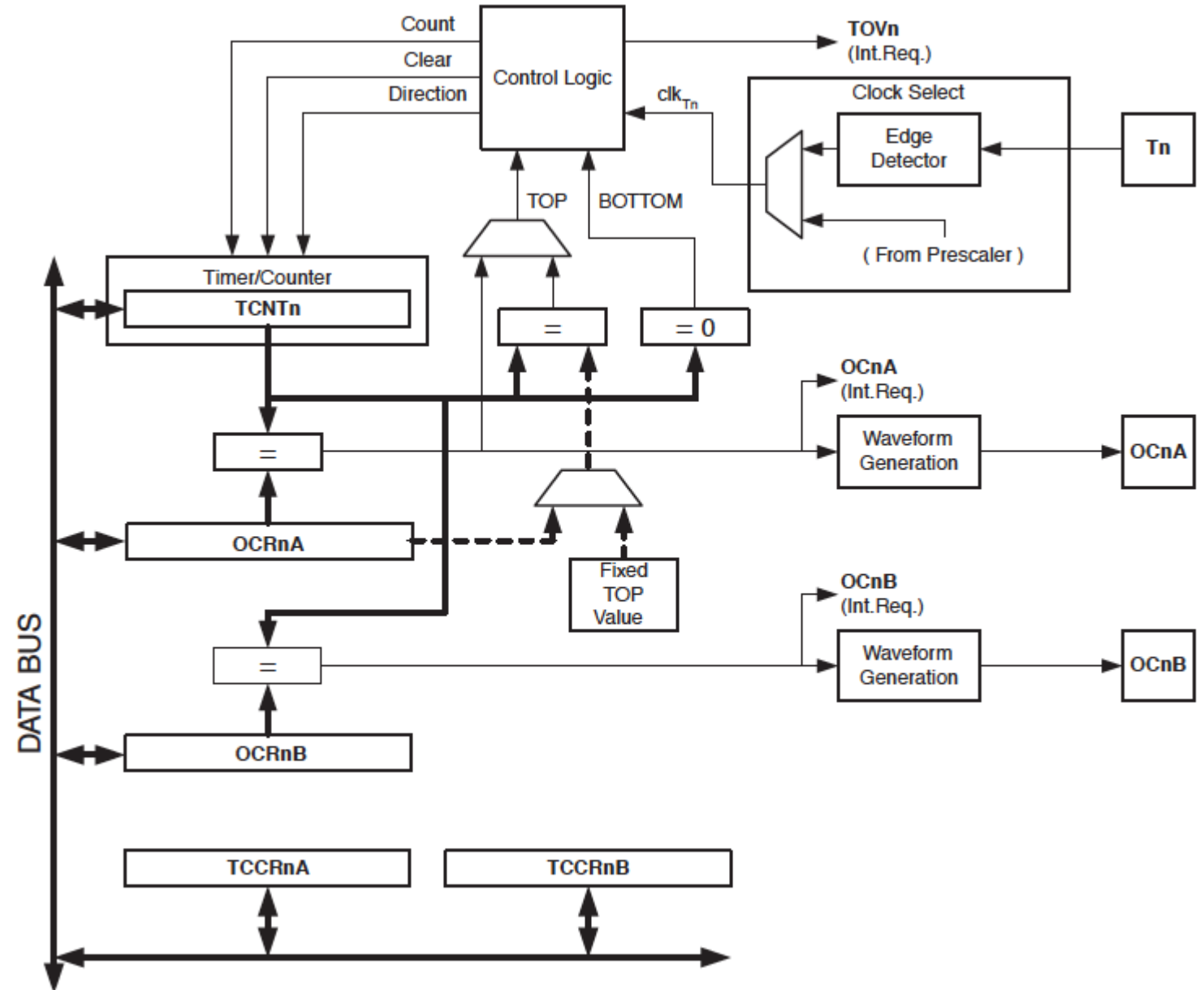


Vježba 11

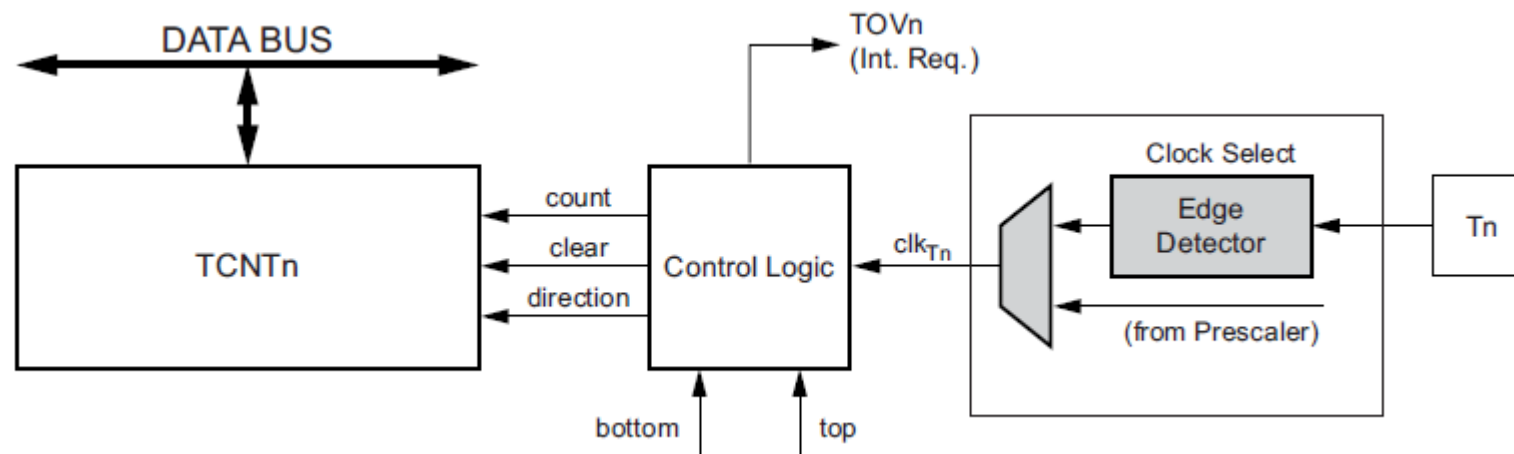
Vremensko –brojački
podsustav

Timer-counter



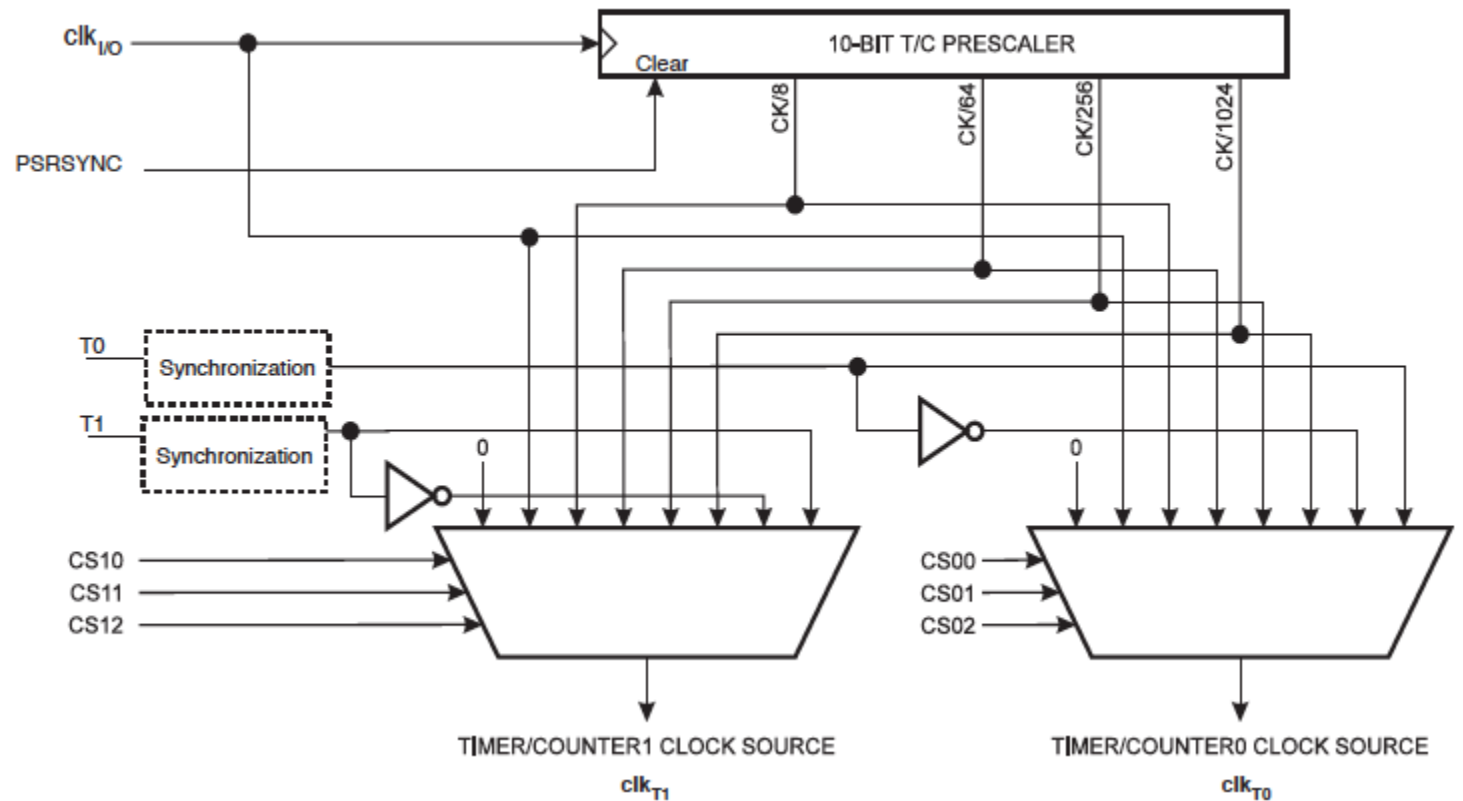
Kontrolna logika (signali)

- **Count** - povećava ili smanjuje TCNT0 za 1.
- **Direction** - određuje smjer brojanja (povećanje ili smanjenje).
- **Clear** – briše TCNT0 registar (postavlja sve bitove u 0).
- **ClkTn** - Timer/Counter takt
- **Top** – Signalizira da je registar TCNT0 dosegao maksimalnu vrijednost.
- **Bottom** - Signalizira da je registar TCNT0 dosegao minimalnu vrijednost (zero).



- Odabir frekvencije signala (“brzine brojanja”)

Figure 16-2. Prescaler for Timer/Counter0 and Timer/Counter1⁽¹⁾



Odabir izvora izvora takta

CS02	CS01	CS00	Description
0	0	0	No clock source (Timer/Counter stopped)
0	0	1	$\text{clk}_{\text{IO}} / (\text{No prescaling})$
0	1	0	$\text{clk}_{\text{IO}} / 8$ (From prescaler)
0	1	1	$\text{clk}_{\text{IO}} / 64$ (From prescaler)
1	0	0	$\text{clk}_{\text{IO}} / 256$ (From prescaler)
1	0	1	$\text{clk}_{\text{IO}} / 1024$ (From prescaler)
1	1	0	External clock source on T0 pin. Clock on falling edge.
1	1	1	External clock source on T0 pin. Clock on rising edge.

- T0 pin – PORT D[4]
- T1 pin – PORT D[5]

Režimi rada vremensko-brojačkog sustava

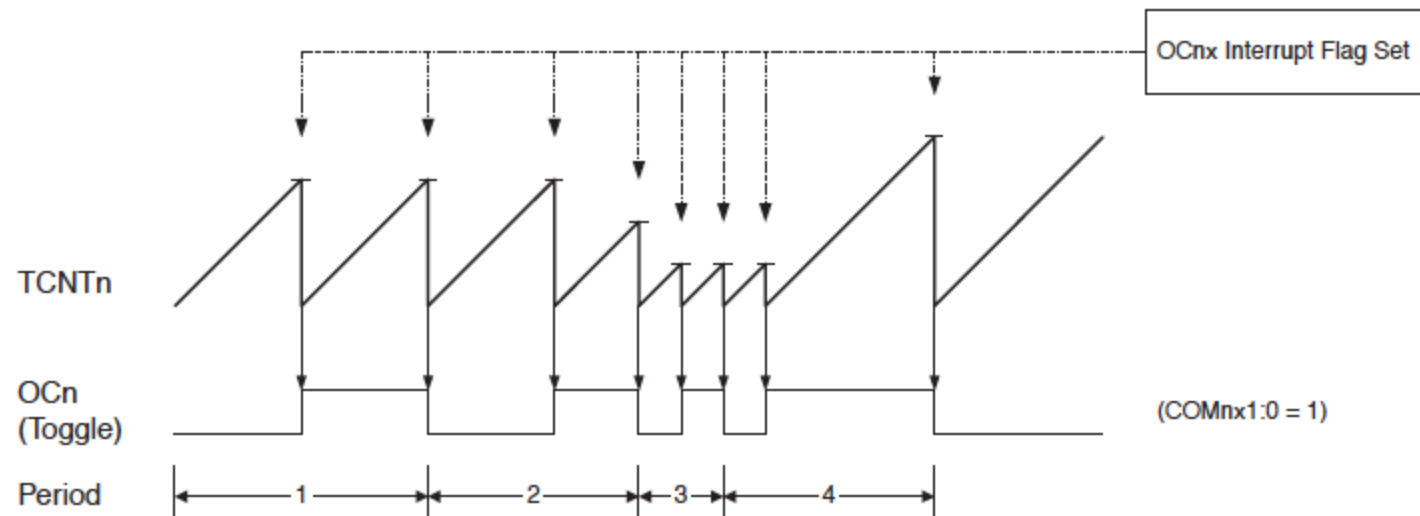
- Normal mode
- CTC (Clear Timer on Compare) mode
- Fast PWM mode
- Phase Correct PWM mode

Normal mode (0)

- Povećava registar za 1
- Maksimalna vrijednost TOP = 0xFF
- Kod ponovnog pokretanja počinje opet s 0
- TOV0 prekidna zastavica se postavlja kad se TCNT0 resetira na 0
- Koristi se za generiranje prekida
 - svaki put kad brojač dosegne vrijednost N (0 - N)
 - Za određeni N broj vremenskih jedinica (255 - N)

CTC (Clear Timer on Compare) Mode (2)

- Brojač povećava vrijednost
- Vrijednosti od 0 do vrijednosti registra OCR0A
 - OCR0A → definira maksimalnu vrijednost brojača
- Kod reseta kreće ponovno od 0
- OCF0A prekidna zastavica se postavlja prilikom reseta (kad brojač ponovno postavlja vrijednost 0)



CTC (Clear Timer on Compare) Mode (2)

- Primjer

Pomoću CTC moguće je generirati signale različitih frekvencija

To se ostvaruje na izlazu OC0A → mijenja vrijednost svaki put kad dosegne postavljenu vrijednost registra OCR0A (do koje broji)

Maksimalna frekvencija: $f_{OC0} = \frac{f_{clk_I/O}}{2}$

Izračun frekvencije osciliranja: $f_{OC0} = \frac{f_{clk_I/O}}{2 * N * (1 + OCRnx)}$

TCCR0A – Timer/Counter Control Register A

Bit	7	6	5	4	3	2	1	0	
0x24 (0x44)	COM0A1	COM0A0	COM0B1	COM0B0	–	–	WGM01	WGM00	TCCR0A
Read/Write	R/W	R/W	R/W	R/W	R	R	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Table 14-2. Compare Output Mode, non-PWM Mode

COM0A1	COM0A0	Description
0	0	Normal port operation, OC0A disconnected.
0	1	Toggle OC0A on compare match
1	0	Clear OC0A on compare match
1	1	Set OC0A on compare match

Table 14-3. Compare Output Mode, Fast PWM Mode⁽¹⁾

COM0A1	COM0A0	Description
0	0	Normal port operation, OC0A disconnected.
0	1	WGM02 = 0: Normal port operation, OC0A disconnected. WGM02 = 1: Toggle OC0A on compare match.
1	0	Clear OC0A on compare match, set OC0A at BOTTOM, (non-inverting mode).
1	1	Set OC0A on compare match, clear OC0A at BOTTOM, (inverting mode).

TCCR0A – Timer/Counter Control Register A

Bit	7	6	5	4	3	2	1	0	
0x24 (0x44)	COM0A1	COM0A0	COM0B1	COM0B0	–	–	WGM01	WGM00	TCCR0A ³
Read/Write	R/W	R/W	R/W	R/W	R	R	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Table 14-5. Compare Output Mode, non-PWM Mode

COM0B1	COM0B0	Description
0	0	Normal port operation, OC0B disconnected.
0	1	Toggle OC0B on compare match
1	0	Clear OC0B on compare match
1	1	Set OC0B on compare match

Table 14-6. Compare Output Mode, Fast PWM Mode⁽¹⁾

COM0B1	COM0B0	Description
0	0	Normal port operation, OC0B disconnected.
0	1	Reserved
1	0	Clear OC0B on compare match, set OC0B at BOTTOM, (non-inverting mode)
1	1	Set OC0B on compare match, clear OC0B at BOTTOM, (inverting mode).

TCCR0A – Timer/Counter Control Register A

Bit	7	6	5	4	3	2	1	0	
0x24 (0x44)	COM0A1	COM0A0	COM0B1	COM0B0	–	–	WGM01	WGM00	TCCR0A
Read/Write	R/W	R/W	R/W	R/W	R	R	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Table 14-8. Waveform Generation Mode Bit Description

Mode	WGM02	WGM01	WGM00	Timer/Counter Mode of Operation	TOP	Update of OCRx at	TOV Flag Set on ⁽¹⁾⁽²⁾
0	0	0	0	Normal	0xFF	Immediate	MAX
1	0	0	1	PWM, phase correct	0xFF	TOP	BOTTOM
2	0	1	0	CTC	OCRA	Immediate	MAX
3	0	1	1	Fast PWM	0xFF	BOTTOM	MAX
4	1	0	0	Reserved	–	–	–
5	1	0	1	PWM, phase correct	OCRA	TOP	BOTTOM
6	1	1	0	Reserved	–	–	–
7	1	1	1	Fast PWM	OCRA	BOTTOM	TOP

TCCR0B – Timer/Counter Control Register B

Bit	7	6	5	4	3	2	1	0	
0x25 (0x45)	FOC0A	FOC0B	–	–	WGM02	CS02	CS01	CS00	TCCR0B
Read/Write	W	W	R	R	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Table 14-8. Waveform Generation Mode Bit Description

Mode	WGM02	WGM01	WGM00	Timer/Counter Mode of Operation	TOP	Update of OCRx at	TOV Flag Set on ⁽¹⁾⁽²⁾
0	0	0	0	Normal	0xFF	Immediate	MAX
1	0	0	1	PWM, phase correct	0xFF	TOP	BOTTOM
2	0	1	0	CTC	OCRA	Immediate	MAX
3	0	1	1	Fast PWM	0xFF	BOTTOM	MAX
4	1	0	0	Reserved	–	–	–
5	1	0	1	PWM, phase correct	OCRA	TOP	BOTTOM
6	1	1	0	Reserved	–	–	–
7	1	1	1	Fast PWM	OCRA	BOTTOM	TOP

TCCR0B – Timer/Counter Control Register B

Bit	7	6	5	4	3	2	1	0	
0x25 (0x45)	FOC0A	FOC0B	–	–	WGM02	CS02	CS01	CS00	TCCR0B
Read/Write	W	W	R	R	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Table 14-9. Clock Select Bit Description

CS02	CS01	CS00	Description
0	0	0	No clock source (Timer/Counter stopped)
0	0	1	clk _{I/O} /(no prescaling)
0	1	0	clk _{I/O} /8 (from prescaler)
0	1	1	clk _{I/O} /64 (from prescaler)
1	0	0	clk _{I/O} /256 (from prescaler)
1	0	1	clk _{I/O} /1024 (from prescaler)
1	1	0	External clock source on T0 pin. Clock on falling edge.
1	1	1	External clock source on T0 pin. Clock on rising edge.

Ostali registri

TIMSK0 – Timer/Counter Interrupt Mask Register

Bit	7	6	5	4	3	2	1	0	
(0x6E)	–	–	–	–	–	OCIE0B	OCIE0A	TOIE0	TIMSK0
Read/Write	R	R	R	R	R	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

- **OCIE0B: Timer/Counter Output Compare Match B Interrupt Enable**
 - Uz uvjet da je Interrupt bit u SREG aktivan, te u OCIE0B upisana “1”, omogućen je prekid prilikom dosezanja zadane vrijednosti tj. dok je postavljen OCF0B bit u **TIFR0 – Timer/Counter 0 Interrupt Flag Registeru**
- **OCIE0A: Timer/Counter0 Output Compare Match A Interrupt Enable**
 - Uz uvjet da je Interrupt bit u SREG aktivan, te u OCIE0A upisana “1”, omogućen je prekid prilikom dosezanja zadane vrijednosti tj. dok je postavljen OCF0A bit u **TIFR0 – Timer/Counter 0 Interrupt Flag Registeru**
- **TOIE0: Timer/Counter0 Overflow Interrupt Enable**
 - Uz uvjet da je Interrupt bit u SREG aktivan, te u TOIE0 upisana “1”, omogućen je prekid prilikom dosezanja maksimalne vrijednosti (overflow) tj. dok je postavljen TOV0 bit u **TIFR0 – Timer/Counter 0 Interrupt Flag Registeru**

Ostali registri

TIFR0 – Timer/Counter 0 Interrupt Flag Register

Bit	7	6	5	4	3	2	1	0	
0x15 (0x35)	–	–	–	–	–	OCF0B	OCF0A	TOV0	TIFR0
Read/Write	R	R	R	R	R	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

- **OCF0B: Timer/Counter 0 Output Compare B Match Flag**
 - upisana “1” → kada vrijednost brojača dosegne vrijednost OCR0B,
 - Vrijednost se briše nakon izvršavanja prekida
- **OCF0A: Timer/Counter 0 Output Compare A Match Flag**
 - upisana “1” → kada vrijednost brojača dosegne vrijednost OCR0A,
 - Vrijednost se briše nakon izvršavanja prekida
- **TOV0: Timer/Counter0 Overflow Flag**
 - Zastavica se aktivira prilikom “overflow-a” brojača 0

Primjer 1.

- Kako se može koristiti TIMER 0 za generaciju delay funkcije uz upotrebu prekida
- Rezim rada Normal mode
- Sklop pali i gasi LED diodu svaki 0.5 sekundi

```

.include "./m328Pdef.inc"

.def temp = r16
.def overflows = r17
.org 0x0000
rjmp Reset
.org 0x0020
rjmp overflow_handler
Reset:
    ldi temp, 0b00000101
    out TCCR0B, temp    ; CPU freq/1024
    ldi temp, 0b00000001
    sts TIMSK0, temp    ; Timer Overflow Interrupt Enable (TOIE0)
bit
    sei                ; enable global interrupts – isto kao "sbi SREG, I"
    clr temp
    out TCNT0, temp     ; postavlja Timer/Counter na 0
    sbi DDRD, 4         ; postavlja PD4 kao izlaz
;=====
; Main body of program:
blink:
    sbi PORTD, 4        ; pali LED na PD4

```

```

rcall delay            ; 1/2 sekunde delay
cbi PORTD, 4          ; gasi LED na PD4
rcall delay            ; 1/2 sekunde delay
rjmp blink

```

```

delay:
    clr overflows      ; postavlja na 0
sec_count:
    cpi overflows,30   ; uspoređuje vrijednost varijable overflows
                        sa 30
    brne sec_count     ; ako je različito vraća se na sec_count
    nop
    ret                ; ako je dosegao vrijednost 30 vraća se iz rutine 3
overflow_handler:
    inc overflows      ; dodaje vrijednost 1 varijabli overflows
    cpi overflows, 61   ; uspoređuje vrijednost varijable overflows
                        sa 61
    brne PC+2          ; ako je različito povećava programsko brojilo
                        za 2 (preskače "clr overflows")
    clr overflows      ; ako je dosegao vrijednost 61 resitira
                        vrijednost
    reti               ; povratak s prekida

```

Upotreba timera/prekida - Arduino Uno

- Odabir vanjskog izvora takta (npr. 2 Hz preko timer1)
 - CPU 16MHz
 - Maximalna vrijednost registra (8bit $\rightarrow$ 256, 16bit $\rightarrow$ 65536)
 - Takt CPU frekvencije dijeli se sa vrijednosti prescalara ($16000000/256=62500$)
 - Podijeliti rezultat s željenom frekvencijom ($62500/2\text{Hz}=31250$)
 - Provjeriti dali vrijednost do koje je potrebno brojiti manja od maksimalne vrijednosti ($31250 < 65536$)